



SOI220 TEST CHIP

Designed by Dr Shengqi Yin

No.	Component list (cascading cells)
1	RIB 1x2 MMI Oband
2	RIB 2x2 MMI Oband
3	Strip 1X4 MMI Cband
4	Strip 2X4 MMI Cband
5	Crossing
6	Y-splitter

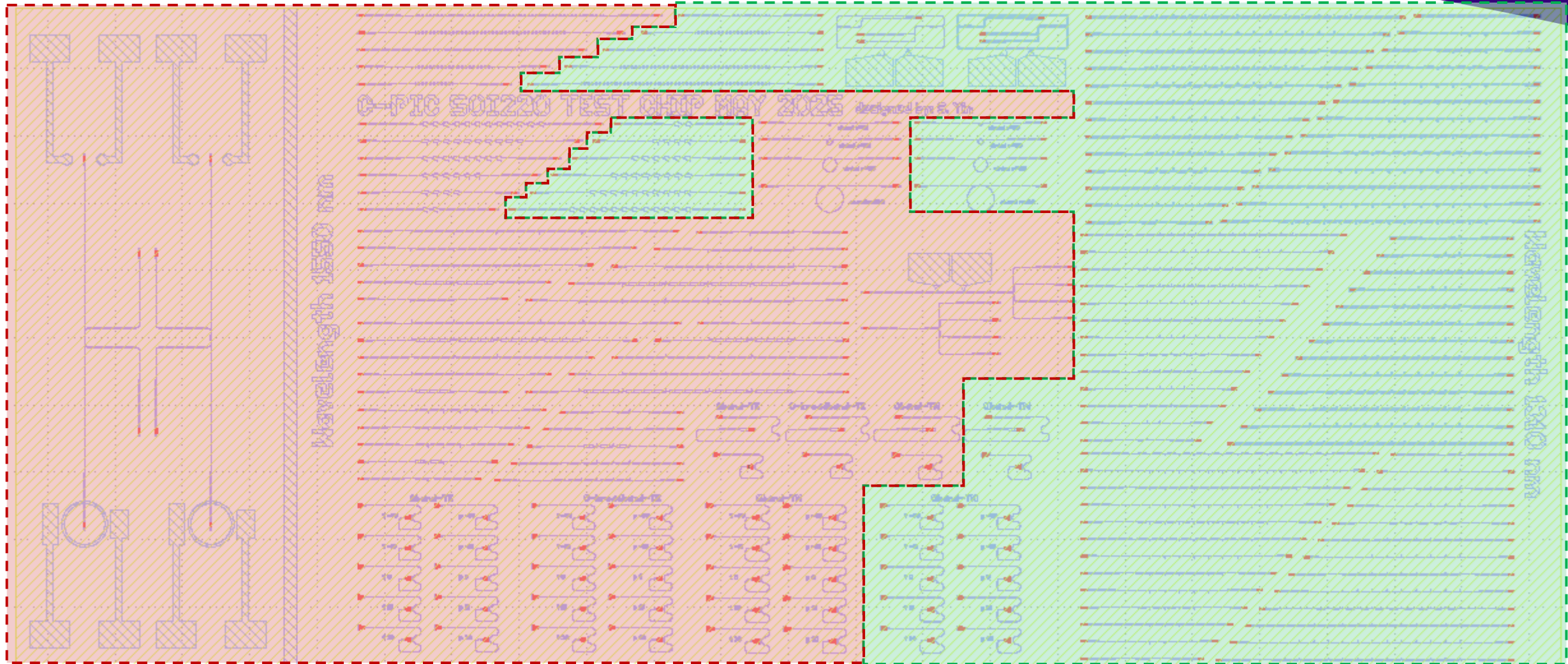
No.	Component test unit
1	1X4 MMI
2	2X4 MMI
3	Mach-Zehnder interferometer
4	Apodized grating coupler
5	Photon detector packaging

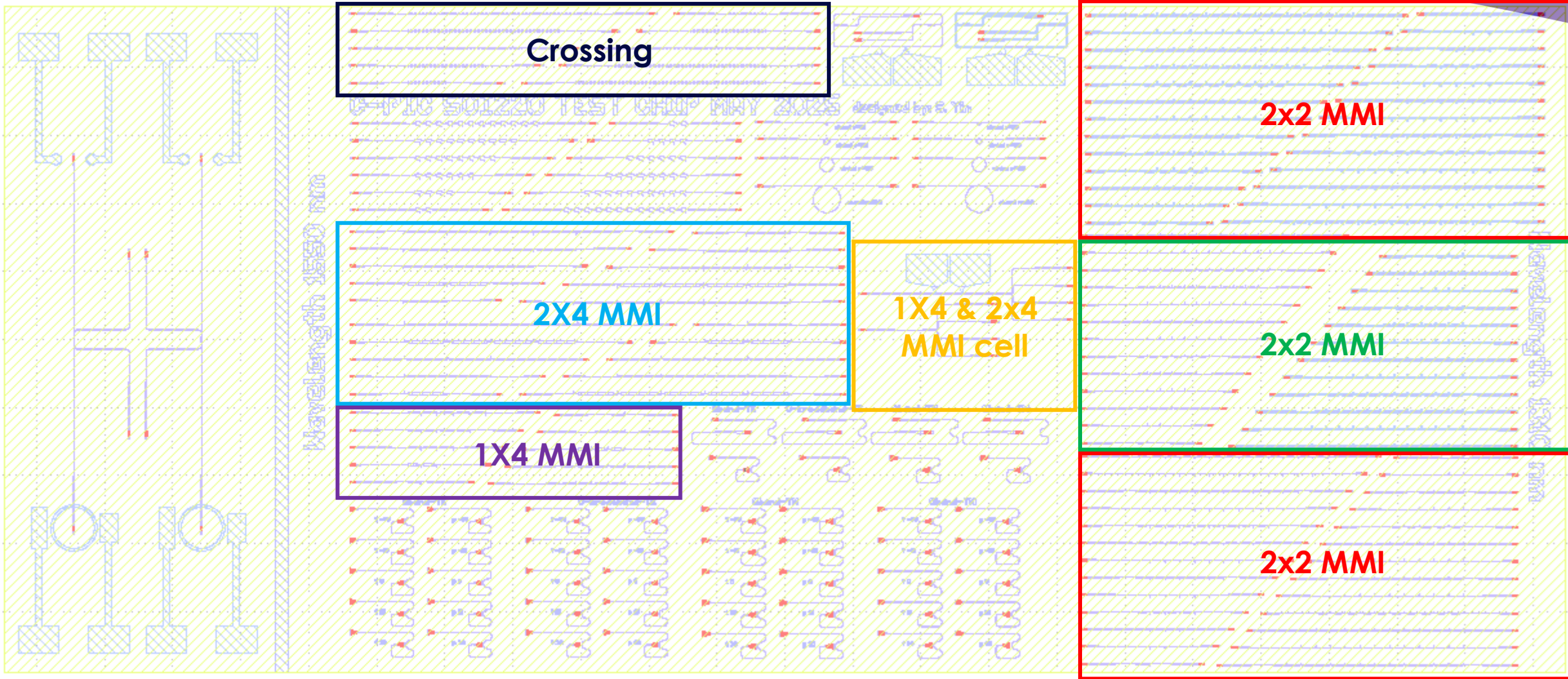
No.	Parameter sweep group
1	Ring Resonator
2	Apodized grating coupler

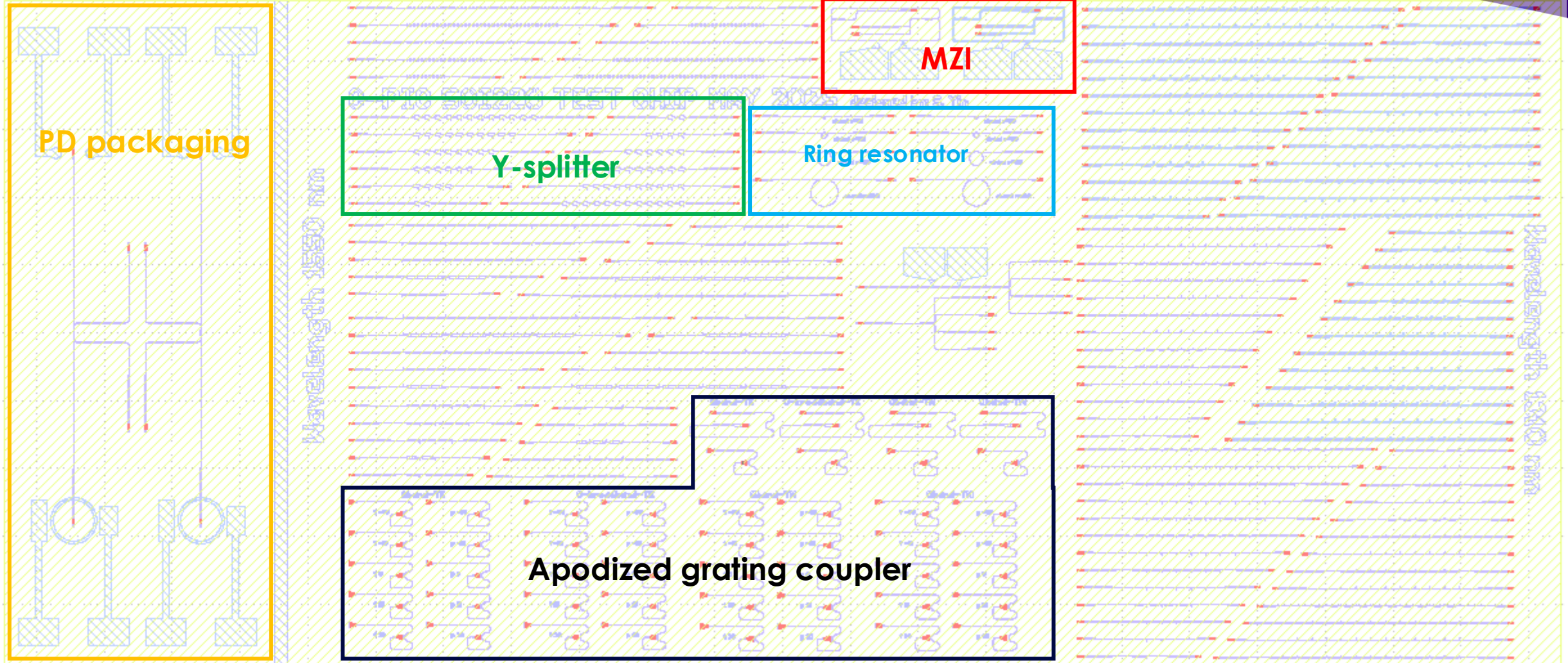
May 2025 MPW #45 – bands distribution



- Red region is designed for 1550 nm (Cband)
- Green region is designed for 1310 nm (Oband)

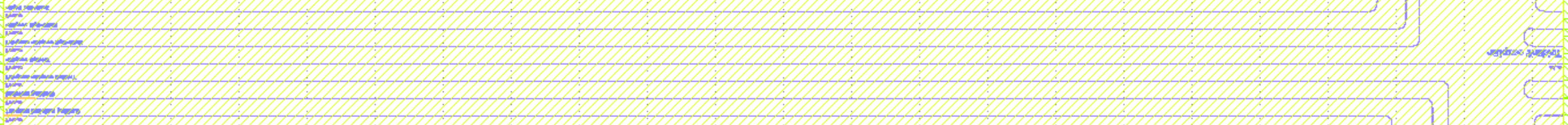




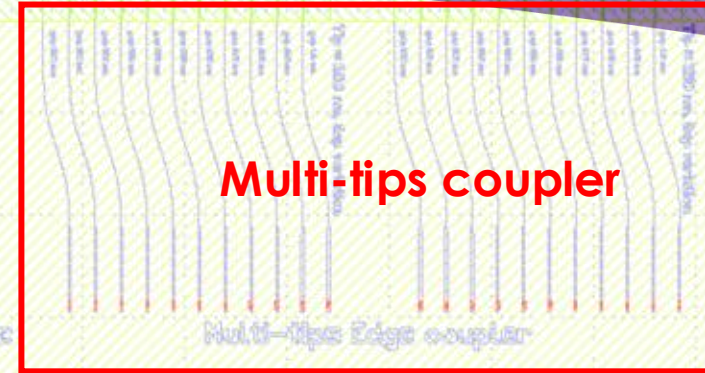
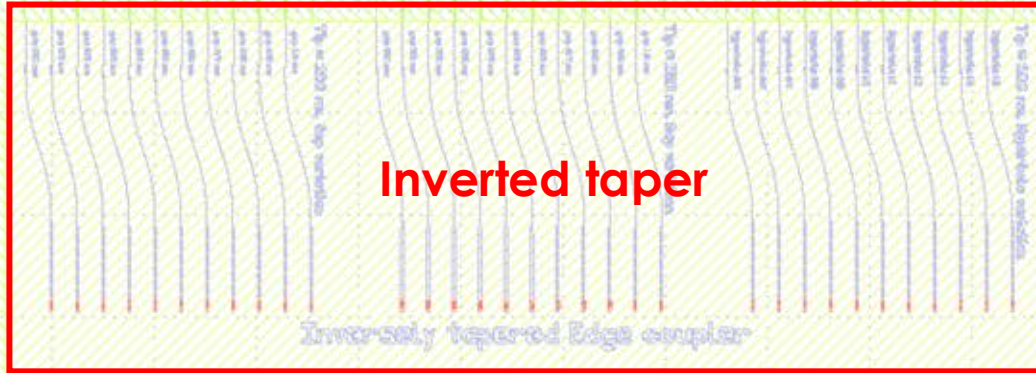


No.	Parameter sweep group
1a	Tip-to-facet gap
1b	Buffer length
2	Hyperbolic coefficient

No.	Component test unit
1	Inversely tapered edge coupler
2	Multi-tips edge coupler
3	Compact Multi-tips edge coupler
4	Trident edge coupler
5	Compact Trident edge coupler
6	Cladding matched edge coupler
7	Compact cladding matched edge coupler

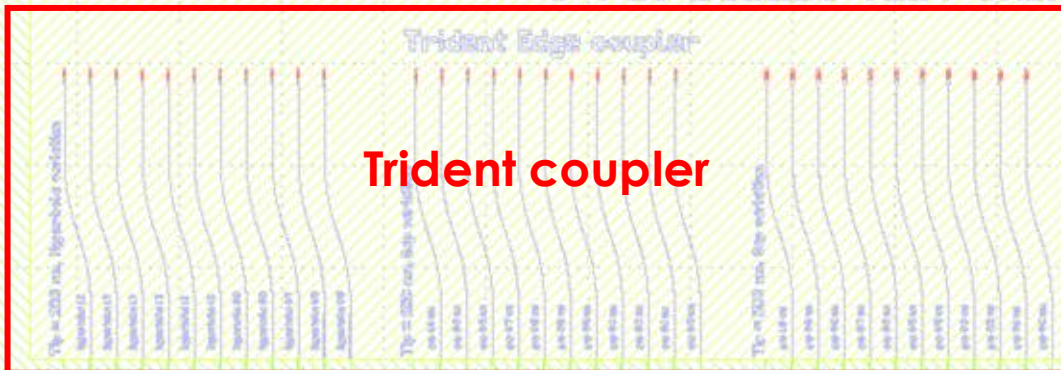


May 2025 MPW #45 type2



Reference

Multi-tips Edge coupler



Trident coupler



Cladding matched coupler

G-PIC 501220 TEST CHIP MAY 2025 - EDGE COUPLERS designed by S. Th

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2	RIB 2x2 MMI Oband
3	Strip 1X4 MMI Cband
4	Strip 2X4 MMI Cband
5	Crossing
6	Y-splitter

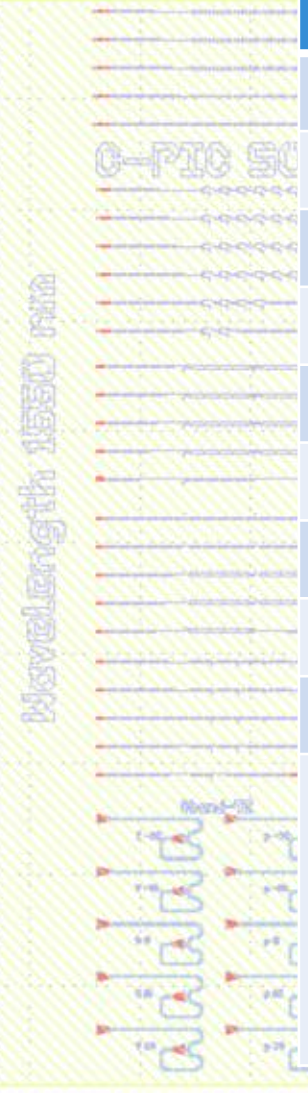
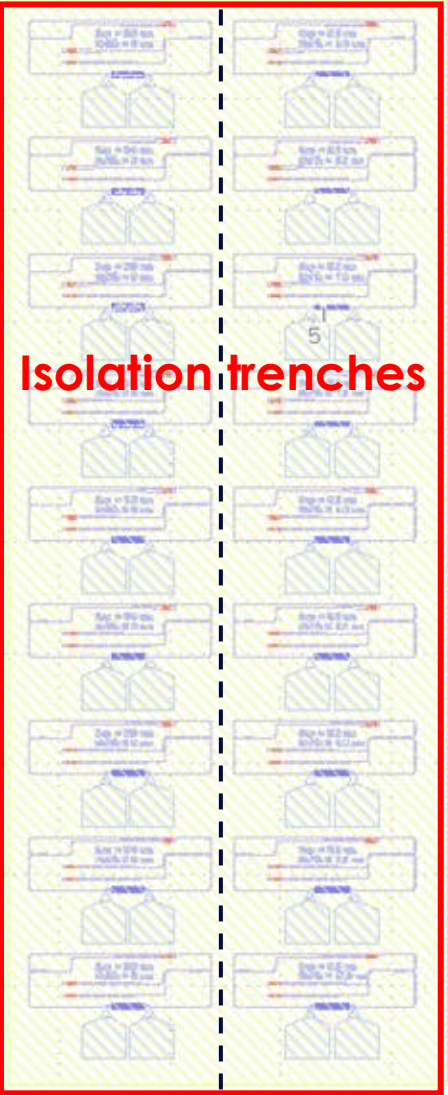
No.	Component test unit
1	1X4 MMI
2	2X4 MMI
3	Mach-Zehnder interferometer
4	Apodized grating coupler - rib

No.	Parameter sweep group
1	Ring Resonator
2	Apodized grating coupler - rib
3	Isolation trench

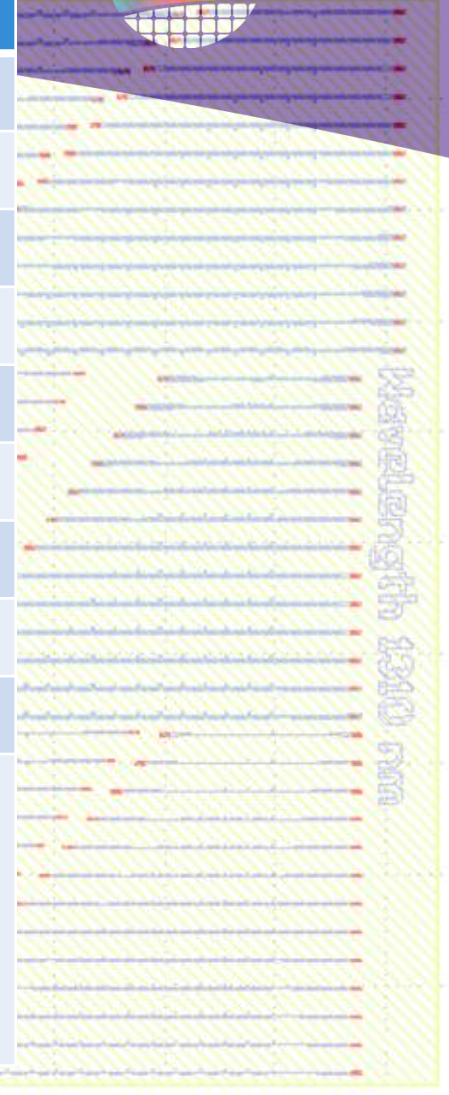
June 2025 MPW #45 type3



Col #1 | Col #2



Col #1 width (fixed) = 9 μ m	Col #2 gap (fixed) = 6.5 μ m
Gap = 5.5 μ m	Width = 6 μ m
Gap = 6 μ m	Width = 6.5 μ m
Gap = 6.5 μ m	Width = 7 μ m
Gap = 7 μ m	Width = 7.5 μ m
Gap = 7.5 μ m	Width = 8 μ m
Gap = 8 μ m	Width = 8.5 μ m
Gap = 8.5 μ m	Width = 9 μ m
Gap = 9 μ m	Width = 9.5 μ m
Gap = 9.5 μ m	Width = 10 μ m
Heater filament with 200- μ m-length and 5- μ m-width Length of isolation trench is 150 μ m for all test cells	



CORNERSTONE

